Combining Repeat Viewing, Personalized PageRank and Association Rules for SWEET.TV: An Evidence-Based LightGBM Top-5 Recommender

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Repository: https://github.com/abaidullahseikh/RecSys-LLMs/tree/main/Exam%20FKN/SWEETTV_Final_Exam

Abstract

The problem is to rank five scheduled TV programme IDs for each of 1,260 SWEET.TV users from channel-viewing intervals, evaluated by MAP@5. I join view intervals to labelled airings and compare popularity, personal frequency, Personalized PageRank (PPR), association rules and content-based filtering (CBF) on a chronological hold-out before considering LightGBM LambdaRank. The Kaggle screenshot shows V10 private MAP@5 = 0.40637 (public 0.42051), the best private score among the four visible submissions. Repeat viewing and co-viewing provide useful signals, but the records also expose missing programme labels and limited recall for unseen titles.

Index Terms — recommender systems, implicit feedback, temporal validation, Personalized PageRank, MAP@5.

1. Introduction

Problem statement. A channel-viewing log must be mapped to TV programme airings by channel equality and interval overlap; the prediction is a ranked list of five positive programme IDs for each of 1,260 users in Weeks 31–42. **Motivation.** Only 10.8% of early schedule rows have a nonzero programme ID, and repeat viewing makes a simple popularity or own-frequency baseline hard to beat. The 120-second local viewing proxy is not equivalent to duration-normalised competition labels. **Concrete example.** For heavy user 13571199046623737884, personal-frequency recommendations match later viewing at ranks 1–4 but not 5, giving AP@5 = 0.800. **Contributions.** A chronological comparison of four approved/related methods plus baselines; a separate V10 LambdaRank contest submission; and documented model, CSV and hand-calculation checks.

2. Related Work

Prior work. Hu et al. [2] motivate implicit-feedback frequency and confidence; Haveliwala [5] describes personalised graph walks; Agrawal and Srikant [6] formalise association rules; Lops et al. [3] cover metadata-driven content filtering; Ke et al. [7] introduce LightGBM. **Alternatives considered.** A popularity-only system scored 0.768418 versus PPR 0.815151 on the local all-distinct target. Strengthening the rule lift threshold to 1.2 lowered AR to 0.707262 versus 0.771331 at the selected setting; CBF scored 0.495029. A V9 unseen-item score correction selected zero, and V10 did not beat V8 on the saved development cohort (0.408498 versus 0.409652). These are documented offline comparisons, not evidence of an online improvement.

3. Method

Approach and pipeline. 1) Parse views and scheduled airings with the inferred Europe/Kyiv clock; overlap = max(0, min(ends) − max(starts)), requiring the same channel. Union fragmented overlaps before the 120-second cutoff; discard tv_show_id = 0 as an item. 2) Train on Weeks 11–20 and validate on Weeks 21–30; final predictions use all history through Week 30 and only future schedule availability. 3) Restrict the candidate universe to labelled programmes scheduled in the target window; allow repeats, rank per method and pad with a shared training-popularity fallback. 4) Generate top-50 candidates from five sources and re-rank with V10 LightGBM LambdaRank using 77 historical, co-view, metadata and schedule features. **Design and configuration.** V10: 78 boosting rounds, 15 leaves, learning rate 0.05, min leaf 100, L2 = 5, truncation 8, seed 2026, four threads; tie-break by popularity then numeric ID. Recorded environments: Python 3.14.4 (early pipeline) and Python 3.12.14, NumPy 2.2.6, SciPy 1.15.3, LightGBM 4.6.0 (ranker).

4. Experiments

The common standalone evaluation uses the 1,260 template users, weeks 11–20 as history and weeks 21–30 as outcomes. The primary local relevance set contains all distinct later programmes with a qualifying view; a secondary sensitivity target keeps each user's later frequent Top-5. The table reports stored local MAP@5; the reported Kaggle private score is separate. The experiments preserve chronological order, schedule-limited candidates, repeated-item permission and a shared train-derived popularity fallback. The official score must not be inferred from the 120-second local proxy.

Input / catalogue	Rows (specification)	Purpose
Viewing intervals	5,821,987	User/channel/time observations
Schedule weeks 11–20	1,074,235	Historical broadcasts and attributes
Schedule weeks 21–30	1,074,394	Validation broadcasts and item pool
Schedule weeks 31–42	1,295,378	Future test broadcasting slots
Submission template	1,260 users	Exactly five programme IDs each

Table 1. Data accounting specified in the exam quiz instructions.

Method	All distinct MAP@5	Frequent Top-5 MAP@5
Popularity	0.768418	0.306627
Personal frequency	0.787627	0.434132
Personalized PageRank	0.815151	0.429489
Association rules	0.771331	0.313643
Content-based filtering	0.495029	0.200193

Table 2. Standalone local MAP@5 on the same 1,260 users. The two metric columns use different local relevance definitions.

The V10 competition model must be interpreted separately. The historical notes report user-cross-fitted Top-5 results of 0.408498 on 3,901 development users and 0.409136 on 992 assessment users, while V8 development scored 0.409652. These V10 cohorts and label protocols differ from the standalone table, so subtracting the scores is invalid. The choice of V10 is supported by the best visible Kaggle private score, not a demonstrated same-cohort V10 win. The two-user strict-temporal proxy comparison is shown below, but a whole-cohort V10-versus-standalones result is not recorded.

Kaggle result and screenshot evidence

The uploaded Kaggle submissions screenshot confirms that final_submission_v10_experimental.csv has the best visible private score (0.40637), while v11 has a slightly lower private score (0.40634) despite a higher public score. This distinction matters because the exam grades the private score.

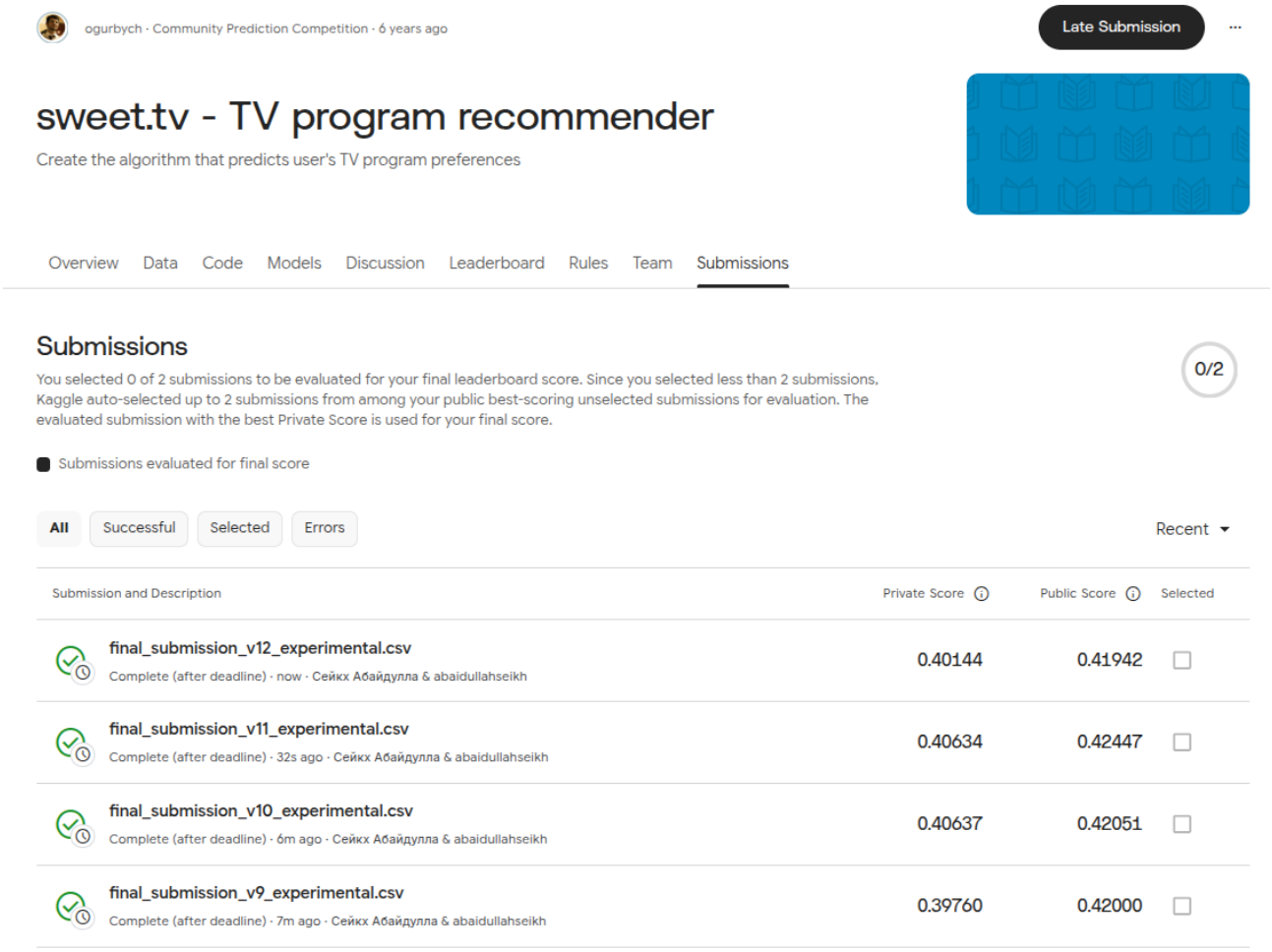


Fig. 1. Kaggle submissions screenshot. Among the visible late submissions, the best private score is 0.40637 for final_submission_v10_experimental.csv.

Repository and submission verification. The public Exam FKN/SWEETTV_Final_Exam folder contains PPR, rules, CBF and LightGBM code and tests. Its submission/ folder now tracks final_submission_v10_experimental.csv with Git LFS. I independently parsed the uploaded 74,959-byte CSV: 1,260 rows and distinct user IDs; each row has five distinct positive programme IDs (6,300 ranked slots), with no malformed rows. SHA-256 39ebef90f28783caa7960bdda8c28347116d7837b3af4fd73c6a9891ea49fd7e matches the Git LFS pointer. The original template user order and test-catalogue eligibility were not rechecked against the unavailable source files; the repository README remains outdated.

Verification and reproducibility. The published deterministic interval fixtures provide checkable counterexamples: the same-channel view [30,180) overlapping an airing [0,100) contributes 70 seconds; a different channel contributes zero. Three overlapping view fragments [10,100), [50,180), [60,70) cover 170 unique seconds, not the naive sum of 230. The repository test oracle enumerates elapsed seconds independently of the optimized join. Recorded Step 2 tests, Step 6 rule checks and V10 checks were agent-run; the entire pipeline was not re-executed for this report.

Check / source	Recorded or directly inspected outcome
Step 2: test_data_pipeline.py	100 seeded brute-force join cases in code; 26 unit tests and 100 full-data checks reported in README
Step 6: association rules	147 checks, zero failures (saved Step 6 verification note)
V10: verify_v10.py	38 final checks, zero failures; exact original CSV replay reported in session
Here: uploaded V10 CSV	1,260 unique user rows; 5 distinct positive IDs each; SHA-256 39ebef90…ea49fd7e

Recommendation examples (same two users across methods)

The examples use the same temporal hold-out and the same fallback policy across methods. An asterisk (*) marks a later validation hit in the preserved evidence. Titles and categories are taken from the saved schedule notes.

Heavy viewer: 13571199046623737884

Selected by largest historical distinct-programme count in the fixed 1,260-user cohort: 757 programmes, 2,358 qualifying airings, and 333 later distinct relevant programmes.

Method / AP@5	Ordered Top-5: programme ID, title [category]
Popularity AP@5 1.000	* 1. 2400480 ТСН. [Инфо]; * 2. 240081 Факты. Утро. [Инфо]; * 3. 2400508 Сегодня. [Инфо]; * 4. 6500479 'Мир наизнанку. Китай', 16 эп. [Познавательное]; * 5. 2400467 'Викна-Новости'. [Инфо]
Personal frequency AP@5 0.800	* 1. 12001682 М/с 'Губка Боб Квадратные Штаны'. [Детям]; * 2. 2400480 ТСН. [Инфо]; * 3. 240081 Факты. Утро. [Инфо]; * 4. 2400508 Сегодня. [Инфо]; 5. 20088 'Завтрак с 1+1'. Информационно-развлекательная программа. [Развлечения]
Personalized PageRank AP@5 1.000	* 1. 2400480 ТСН. [Инфо]; * 2. 240081 Факты. Утро. [Инфо]; * 3. 12001682 М/с 'Губка Боб Квадратные Штаны'. [Детям]; * 4. 2400508 Сегодня. [Инфо]; * 5. 6500479 'Мир наизнанку. Китай', 16 эп. [Познавательное]
Association rules AP@5 1.000	* 1. 240081 Факты. Утро. [Инфо]; * 2. 2400480 ТСН. [Инфо]; * 3. 2400508 Сегодня. [Инфо]; * 4. 6500479 'Мир наизнанку. Китай', 16 эп. [Познавательное]; * 5. 2400467 'Викна-Новости'. [Инфо]
Content-based filtering AP@5 0.000	1. 10003148 Т/с 'Стоматолог', 1 с. [Сериалы]; 2. 10003102 Т/с 'Опер по вызову', 2 сезон, 9 с. [Сериалы]; 3. 400471 Заробітчани, 4 сезон, 1 эп. Турция. [Развлечения]; 4. 10003080 Т/с 'Школа', 1 с. [Сериалы]; 5. 1000608 Т/с 'Приключения Геркулеса', 5 сезон, 1 и 2 с. [Сериалы]
V10 temporal proxy AP@5 0.800	* 1. 2400480 ТСН. [Инфо]; * 2. 240081 Факты. Утро. [Инфо]; * 3. 12001682 М/с 'Губка Боб Квадратные Штаны'. [Детям]; * 4. 2400508 Сегодня. [Инфо]; 5. 20088 'Завтрак с 1+1'. Информационно-развлекательная программа. [Развлечения]

Table 3. Heavy viewer: every method’s five ranked IDs, schedule titles, categories and per-method AP@5.

Heavy-user analysis. The first five popularity recommendations, all five PPR recommendations and all five association-rule recommendations appeared in this viewer’s later qualifying programme set (AP@5 = 1.000 each). Personal frequency put “Завтрак с 1+1” at rank 5 without a later qualifying viewing, so it achieved 0.800. The strictly historical V10-configuration proxy also achieved 0.800, while the content-based list had no observed matches. These examples support repeat viewing and graph diffusion for this user; they do not measure satisfaction.

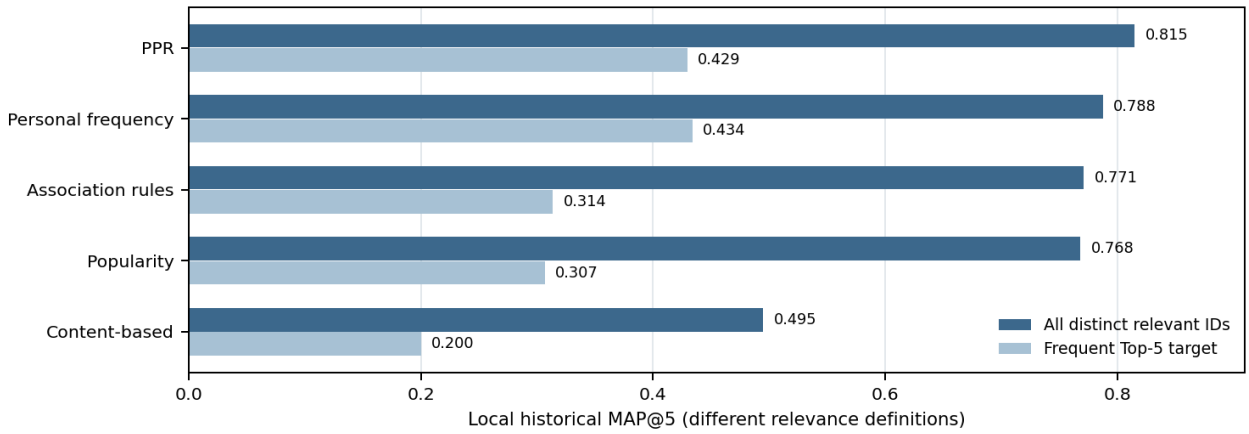


Fig. 2. Stored standalone MAP@5 values, same 1,260 users and chronological hold-out. Each pair of bars uses a different truth definition; the chart is not a V10 hold-out comparison.

Light viewer: 8150816946177670084

Selected by smallest positive historical count in that cohort: one qualifying film (ID 90053098), 2,753 seconds of qualifying exposure, and 15 later distinct relevant programmes.

Method / AP@5	Ordered Top-5: programme ID, title [category]
Popularity AP@5 0.100	1. 2400480 ТСН. [Инфо]; * 2. 240081 Факты. Утро. [Инфо]; 3. 2400508 Сегодня. [Инфо]; 4. 6500479 'Мир наизнанку. Китай', 16 эп. [Познавательное]; 5. 2400467 'Викна-Новости'. [Инфо]
Personal frequency AP@5 0.067	1. 90053098 Х/ф 'Мужество в бою'. [Фильмы]; 2. 2400480 ТСН. [Инфо] (fallback); * 3. 240081 Факты. Утро. [Инфо] (fallback); 4. 2400508 Сегодня. [Инфо] (fallback); 5. 6500479 'Мир наизнанку. Китай', 16 эп. [Познавательное] (fallback)
Personalized PageRank AP@5 0.100	1. 90053098 Х/ф 'Мужество в бою'. [Фильмы]; * 2. 240081 Факты. Утро. [Инфо]; 3. 2400480 ТСН. [Инфо]; 4. 6500479 'Мир наизнанку. Китай', 16 эп. [Познавательное]; 5. 2400467 'Викна-Новости'. [Инфо]
Association rules AP@5 0.000	1. 90075859 Х/ф 'Экипаж'. [Фильмы]; 2. 90081867 Х/ф 'Миссия невыполнима: Последствия'. [Фильмы]; 3. 90077612 Х/ф 'Джек Райан: Теория хаоса'. [Фильмы]; 4. 90057593 Х/ф 'Сокровище Амазонки'. [Фильмы]; 5. 90076096 Х/ф 'После нашей эры'. [Фильмы]
Content-based filtering AP@5 0.000	1. 90053098 Х/ф 'Мужество в бою'. [Фильмы]; 2. 90082673 Премьера. Х/ф 'Кролик Джоджо'. [Фильмы]; 3. 90064040 Х/ф 'Всемирный торговый центр', 1 ч. [Фильмы]; 4. 90045696 Х/ф 'Пуленепробиваемый'. [Фильмы]; 5. 90063567 Х/ф 'Второй в команде'. [Фильмы]
V10 temporal proxy AP@5 0.200	* 1. 240081 Факты. Утро. [Инфо]; 2. 2400480 ТСН. [Инфо]; 3. 2500430 'Затерянный мир'. [Познавательное]; 4. 2400508 Сегодня. [Инфо]; 5. 20088 'Завтрак с 1+1'. Информационно-развлекательная программа. [Развлечения]

Table 4. Light viewer: the same method families and complete Top-5 output.

Why the lists differ. Frequency repeats historically frequent programmes; PPR spreads preference through co-view neighbours and can reorder the heavy viewer's news/entertainment mix. Association rules express inspectable A→B co-view links from 14-day baskets (confidence and lift), whereas matrix factorization would learn latent user/item affinities without explicit directional rules; MF was NOT implemented or compared here. CBF uses category, genre, actors, director and other programme metadata where collaborative co-view evidence is sparse. Yet neither selected user's CBF Top-5 contained a later qualifying hit, so metadata similarity is not proven useful discovery. Popularity achieves five heavy-viewer matches but just one light-viewer hit at rank 2; its shared fallback is reliable but not personal.

Light-user analysis. This viewer has one qualifying historical film (2,753 seconds) and 15 distinct later relevant programmes. The frequency method required four popularity backfill slots. The V10 temporal proxy matched one later programme at rank 1 (AP@5 = 0.200), while popularity and PPR each matched at rank 2 (0.100), frequency matched at rank 3 (0.067) and rules/CBF matched none. Unseen films are not necessarily good discoveries; a missing platform view is not evidence that a programme was never seen elsewhere.

Method	Heavy: hits / AP@5	Light: hits / AP@5
Popularity	5 / 1.000	1 / 0.100
Personal frequency	4 / 0.800	1 / 0.067
PPR	5 / 1.000	1 / 0.100
Association rules	5 / 1.000	0 / 0.000
Content-based	0 / 0.000	0 / 0.000
V10 temporal proxy	4 / 0.800	1 / 0.200

Table 5. AP@5 for the same two users on the 120-second local all-distinct relevance target (not the Kaggle metric). The V10 temporal proxy was trained with Weeks 11–15 features and Weeks 16–20 labels, then evaluated on Weeks 21–30; the production V10 model was fitted using later information and must not be scored on that historical window as if unseen.

5. Discussion — which strategy for SWEET.TV, and why?

Recommendation and operational scope. Choose V10 LambdaRank as the designated competition submission: private MAP@5 = 0.40637 is best among the four visible Kaggle attempts; the available local cohorts do not demonstrate a same-hold-out win over every standalone method. V10 blends personal repetition, PPR, rules, content and schedule candidates, with popularity completing sparse lists. This fits an offline personalized "for you" row while keeping familiar titles available; a separate exploratory row would need its own relevance test. Limitations. Only 10.8% of early schedule airtime has IDs; in a broader V9 audit, 3,445/5,774 relevant unseen labels (59.66%) were absent from the candidate pool, beyond any re-ranker's reach. Local 120-second labels are not Kaggle truth. Missing platform views do not prove a person never watched a show, and unfamiliar items are not automatically useful. No satisfaction, online A/B or revenue benefit was measured. Next: expand candidate recall, then compare all models on one untouched temporal hold-out with duration-normalized sensitivity checks.

Cost and evidence limits. On the recorded four-logical-CPU machine, popularity predicted 1,260 users in 0.006 s, personal frequency in 1.603 s, PPR in 14.665 s, rules in 21.338 s, and CBF in 13.725 s; V10 cached-feature final generation took 22.834 s and its full experiment 90.271 s. These were different offline scopes, NOT matched per-request latency benchmarks. Caching and batch updates are plausible deployment approaches, but serving cost remains untested. Improvement loop. With the cosine graph and k=30 fixed, raising PPR restart probability 0.20→0.35 raised local MAP@5 0.801222→0.815151; V10 development MAP@5 (0.408498) still trailed V8 (0.409652), so the Kaggle decision is not evidence of universal dominance.

6. AI Usage Disclosure

Tools and tasks. OpenCode (session metadata: 1.18.31) assisted requirement analysis, Python implementation, debugging, model experiments and tests. Early pipeline logs report Python 3.14.4; later model logs report Python 3.12.14 and LightGBM 4.6.0. **Personally verified.** I checked the saved heavy-viewer Top-5 flags and manually calculated AP@5 = 0.800; I also confirmed inspecting the submitted CSV. **Assistant verification.** Independently parsed 1,260 uploaded CSV rows with 5 positive distinct programme IDs each and compared SHA-256 against the Git LFS record; it also checked two published interval fixtures. **Limits / session record.** The 890 MB raw inputs were not rerun here, and neither template order nor Kaggle hidden labels were independently checked. The original seikh_abaidullah_exam_session.json remains the unmodified coding-tool audit trail.

Rubric evidence: Top-Down 5-Loop

BUILD: The saved popularity reference achieved local primary MAP@5 = 0.768418. ASK WHY: A 120-second local viewing proxy, repeated viewing and unlabelled airtime can make apparently strong metrics misleading, so alternative definitions and baseline comparisons matter. **TEST**: The same chronological hold-out and candidate rules were used across models; I checked the saved heavy-viewer Top-5 relevance flags and independently computed AP@5 = 0.800; automated OpenCode checks are separate. **EXPLAIN**: Co-view diffusion is useful in the local all-distinct target, but it does not establish programme enjoyment or universal superiority. **IMPROVE**: Holding the cosine graph, frequency restart and 30 neighbours fixed, changing PPR restart probability from 0.20 to 0.35 increased primary MAP@5 from 0.801222 to 0.815151 (+0.013929).

Recorded failure, fix and verification

Failure and correction (observed in OpenCode). The first seen/unseen audit CSV writer stopped because the per-user dictionaries sometimes omitted a counter column; the partial eight-row CSV remained preserved. The code then initialized all four slot/hit output fields before writing, and the resumed audit reported 38 passing V10 final checks. This documents an actual before/after change, not a claim that the student manually reran the entire model.

Personal verification (my manual check)

For heavy user 13571199046623737884, the saved personal-frequency Top-5 matches validation at ranks 1, 2, 3 and 4, but not 5: AP@5 = (1/1 + 2/2 + 3/3 + 4/4 + 0)/5 = 0.800. I checked the saved list and its flags, then calculated the metric from the AP@5 formula. My manual check covers this example; I did not rerun the entire data pipeline.

Independent assistant cross-check of public fixtures

Additional assistant cross-check of published fixtures: a view [30,180) and same-channel airing [0,100) overlap for min(180,100)-max(30,0) = 70 seconds; another channel does not match. Fragments [10,100), [50,180), [60,70) together cover [10,180), giving 170 unique watched seconds rather than 230 seconds obtained by naive addition.

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